



JUICE interplanetary operations design drivers and challenges

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I. Nomenclature

<i>ADS</i>	=	Airbus Defence and Space	<i>LEOP</i>	=	Launch and Early Orbit Phase
<i>AU</i>	=	Astronomic Unit	<i>MGA</i>	=	Medium Gain Antenna
<i>CFDP</i>	=	CCSDS File Delivery Protocol	<i>NAVCAM</i>	=	Navigation Camera
<i>CDMU</i>	=	Command and Data Management Unit	<i>OBC</i>	=	On-Board Computer
<i>ESA</i>	=	European Space Agency	<i>OH</i>	=	Optical Head
<i>ESOC</i>	=	European Space Operations Center	<i>PCDU</i>	=	Power Control and Distribution Unit
<i>FDIR</i>	=	Failure Detection, Isolation and Recovery	<i>PDU</i>	=	Protocol Data Unit
<i>GOI</i>	=	Ganymede Orbit Insertion	<i>PUS</i>	=	Packet Utilization Standard
<i>HGA</i>	=	High Gain Antenna	<i>RIU</i>	=	Remote Interface Unit
<i>JOI</i>	=	Jupiter Orbit Insertion	<i>SGM</i>	=	Safe Guard Memory
<i>JUICE</i>	=	JUpiter ICy moons Explorer	<i>SpW</i>	=	SpaceWire
<i>LA</i>	=	Logical Address	<i>SSMM</i>	=	Spacecraft Science Mass Memory
<i>LGA</i>	=	Low Gain Antenna	<i>STR</i>	=	Star Tracker

II. Introduction

JUICE is an European Space Agency programme aiming at sending a spacecraft to observe Jupiter and its moons (Callisto, Ganymede and Europa). JUICE spacecraft has entered in development phase in July 2015 and has successfully passed its preliminary design review in March 2017.

This paper provides an overview of JUICE mission and its scientific goals. It describes the spacecraft journey from Earth to Jupiter: a seven year interplanetary cruise from its launch in June 2022 to its arrival in Jupiter system in 2029, made of several planetary swing-bys (Venus, Earth and Mars), and followed by a 3-years Jupiter tour made of 29 moon fly-bys culminating in a first-ever orbital capture around Jupiter's largest moon Ganymede.

JUICE spacecraft design is driven by the presence of 10 instruments connected to a 1Tbit mass memory through a SpaceWire network, by demanding EMC cleanliness and pointing requirements (to meet scientific objectives), several tons of propellant, and a 85m² solar generator. The mission itself presents a number of challenges due to long range communications, extreme temperature variations, harsh radiation environment and limited power at Jupiter, time-critical manoeuvres such as Jupiter Orbit Insertion (JOI), Ganymede Orbit Insertion (GOI) and Ganymede orbit circularization.

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III. A long journey from Earth to Jupiter

A. From Earth to Jupiter

JUICE trajectory is designed by ESOC Mission Analysis team. It is the result of an optimization, iterated with ESA project and Airbus teams, aiming at maximizing the dry mass at launch, keeping the total mass within the launcher capabilities and minimizing the transfer duration, while respecting other mission constraints, like total radiation dose, minimum Sun distance and maximum eclipse duration.

JUICE spacecraft launch window opens on 2022, May 20th, by ARIANE V, from Kourou. The launcher injects the spacecraft in a direct escape trajectory. About 1 year later, JUICE executes its first Gravity Assistance Manoeuvre; flying at an altitude of 12700 km above the Earth at the closest approach and takes the direction of Venus. After a second Gravity Assistance Manoeuvre around Venus (closest approach altitude of 9500 km), JUICE flies back to Earth for the third Gravity Assistance Manoeuvre (closest approach altitude of 1900 km), then to Mars (closest approach altitude of 1100 km) and then again back to Earth (closest approach altitude of 3600 km) before, eventually, flying to Jupiter.

About 7 years and 4 months after launch, JUICE arrives in the vicinity of Jupiter and has to execute a mission-critical trajectory manoeuvre (the so-called Jupiter Orbit Insertion) to achieve a highly elliptical orbit around Jupiter. A few hours before JOI, a fly-by of Ganymede (closest approach altitude of 400 km) contributes to the overall insertion strategy. Both the fly-by and the JOI aim at reducing the spacecraft velocity to achieve the targeted orbit. 6 month after JOI, a perijove raising manoeuvre is executed close to the apojuve of the orbit.

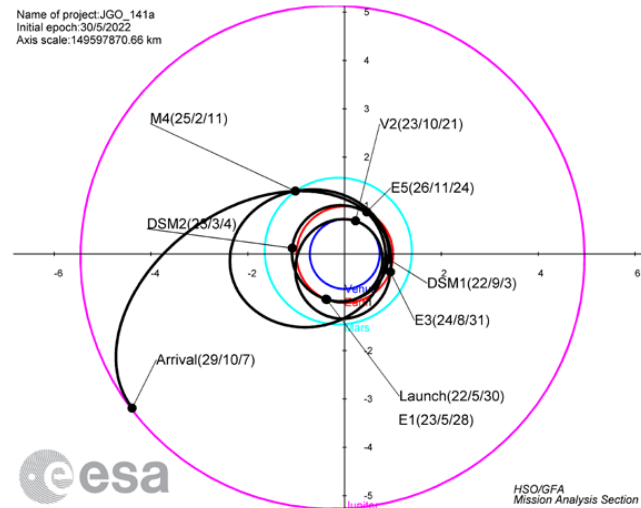


Fig. 1 Baseline scenario for JUICE transfer from Earth to Jupiter

The scenario presented above is not the only one and the mission analysis features a number of back-up trajectories. However, the baseline scenario is one of the most efficient ones in terms of launch mass optimization and duration of the transfer. As an example, in case of a launch in 2023, the transfer would last 10 year instead of 7. For this reason, all the Juice teams (ESA, Industry and science community) are focused to meet the launch rendezvous in May 2022.

The shape of the trajectory is one of the drivers of the spacecraft design: the distance from the Sun varies from 0.64 AU (at closest sun approach) to 5.5 AU leading to a wide range of temperatures. The other main design drivers for Spacecraft and operations design are intrinsically related to Jupiter and do not really depend on the way the spacecraft arrives at Jupiter : (i) distance to earth leading to a 1h20 minutes turn-around time, (ii) Jupiter is away from the Sun leading to limited power resources, (iii) Jupiter is surrounded by a harsh radiative environment requiring special care in the components and materials selection. The criticality of the Jupiter Orbit Insertion manoeuvre is one of the key-driver for the Spacecraft autonomy, leading to implement a brand-new fail-operational concept that allows resuming the critical manoeuvre even if a serious failure would affect one of the spacecraft functions mandatory for mission execution. In other circumstances (non-mission critical phases), such a serious failure would simply lead to a Safe Mode: the usual Fail-Safe concept that would interrupt the mission and wait for ground intervention to resume the mission.

B. The Jupiter tour

During the Jupiter tour, JUICE visits the three Jovian moons Europa, Ganymede and Callisto in a series of more than 20 fly-bys at an altitude down to 400 km. JUICE executes only 2 fly-bys over Europa because the radiation environment does not allow for more. JUICE science objectives include not only the moon observation but also Jupiter observation, which takes place around the perijove of its orbit. A series of Callisto fly-bys allows to increase (and then decrease) the inclination of JUICE orbit to provide a better observability of Jupiter poles.

C. Orbiting Ganymede

After performing a tour of the Galilean moons for about 3 years, JUICE executes another mission-critical manoeuvre to insert into an elliptical orbit around Ganymede. The natural evolution of this orbit leads to a circular orbit (at 5000 km altitude) before becoming elliptical again. A last trajectory correction allows circularizing the orbit around Ganymede at 500 km altitude. This final orbit is a polar orbit. The angle between the orbit plane and the sun direction evolves slowly to provide various lightning conditions for Ganymede observation.

At the end of the mission, JUICE naturally crashes on Ganymede.

IV. Juice spacecraft overview

A. 10 instruments

JUICE spacecraft mission consists in bringing 10 instruments to the Jovian system to observe Jupiter and its moons. JUICE instruments can be divided in 3 categories: (i) remote sensing, (ii) geophysics and (iii) in-situ observations.

The remote sensing instruments are focused on observation of Jupiter and its icy moons, their surfaces and the composition of their atmospheres. This package includes spectro-imaging capabilities from the ultraviolet to the near-infrared, wide angle and narrow angle cameras and a submillimetre wave instrument:

- JANUS is a multispectral imager. From a high altitude, the instrument provides global mapping of the moons and context images around Ganymede, Europa and Callisto. From low altitude, it produces high resolution images for target areas on the icy moons, with spatial resolution from 25 m/pixel down to 2.4 m/pixel.
- MAJIS is a spectrometer operating in the visible and near infra-red bands. It is complementary to the JANUS visible imager. It contributes to the understanding of the Jovian system and its evolution.
- SWI, the Sub-mm Wave Instrument is a two bands spectrometer, concentrating on the Jupiter stratosphere and more particularly on its molecules and vertical winds. It also provides information on the moons' atmospheres.
- UVS is a spectrometer operating in Ultra-Violet bands; it is dedicated to atmosphere and aurora observation of Jupiter and also of its icy moons, which has never been done yet.

The geophysics instruments are dedicated to the restitution of the surface topography, the sub-surface composition and the gravity fields. The geophysical package explores the surface and subsurface of the moons using laser altimetry and radar sounding, complemented by the radio science instruments which probes the Jovian/satellite atmospheres and enables estimation of gravity fields. This set of instruments gathers:

- GALA is a laser altimeter, operating as a bi-static lidar with a 1064 nm wavelength, providing moons' topographic information.
- RIME is a radar, using a 16 m dipole antenna provided by the spacecraft. Thanks to its low frequency, around 9 MHz, it can sound the sub-surface up to few kilometres; it is therefore especially suited to ice body and ocean investigations.
- 3GM is the radio-science instrument. It includes an Ultra-Stable Oscillator (USO), a High Accuracy Accelerometer (HAA) and a Ka-band transponder (KaT) for radio-science. The experiment relies on the spacecraft communications links and provides accurate range and range rate data from which moons' gravity fields are deduced. The high accuracy accelerometer improves 3GM science performances.

The in-situ instruments objective is to provide data on the Jovian environment, mainly on the Jupiter magnetosphere, the plasma and the fields surrounding the moons:

- JMAG is based on two fluxgates and a scalar magnetometer located at the end of a long boom provided by the spacecraft, so as to bring the spacecraft magnetic disturbances under the specified level.
- RPWI, Radio and Plasma Wave Instrument, performs three types of observations to support the Ganymede plasma analysis: plasma waves electrical measurement (LP-PWI), radio emission (RWI) and alternative magnetic field (SCM) measurements.

- PEP includes six sensors spread over four units to cover the 4π sr around the spacecraft. It observes plasma (positive and negative ions, electrons) and neutral atom flows over a wide energy range, with angular and spectral resolution. The aim is to understand the plasma and magnetosphere interactions, and particle weathering.

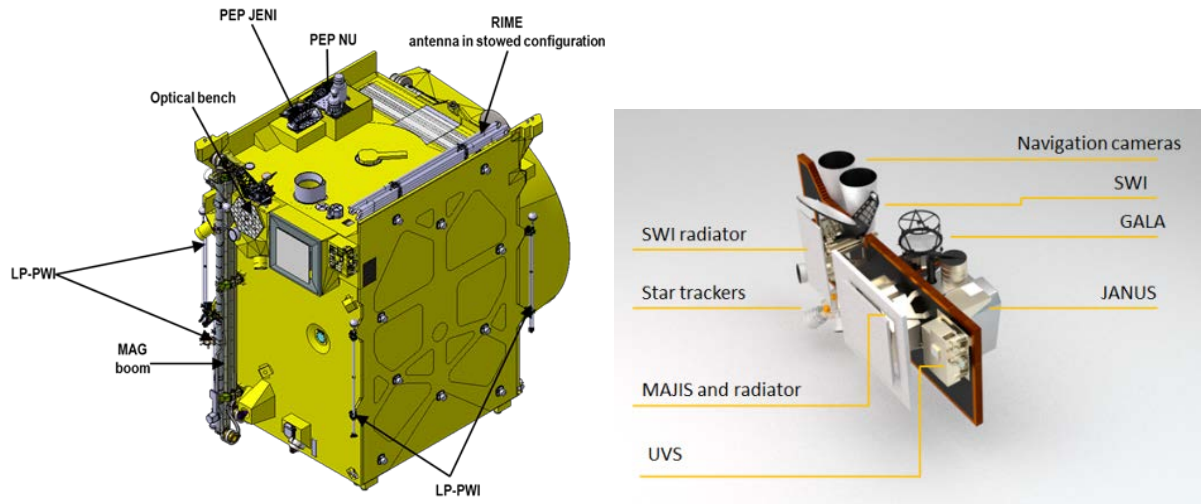


Fig. 2 JUICE instruments accommodation and focus on optical bench

B. Spacecraft accommodation

JUICE spacecraft is based on a structure featuring a central cylinder mounted on a cone to interface with the launcher via a standard adapter. The central cylinder hosts two propellant tanks, six shear walls and four lighten external walls.

JUICE spacecraft is fully dedicated to instruments support, with:

- A nadir face on the spacecraft +Z face opposite to the main engine face. This face supports also PEP-Lo nadir unit and PEP-Hi JENI sensor and the RIME antennas, positioned away from propulsion plumes, minimising thermal loads and contamination from propulsion plumes and residuals.
- An optical bench (Fig. 2) accommodating instruments with stringent pointing requirements close to Star Tracker Optical Heads (STR-OH) and Navigation Cameras. The optical bench is positioned close to one of the vaults in order to ensure limited distances between sensors on the optical bench & related proximity electronics inside the vault.
- A 10.6 m long boom supporting the JMag sensors, RWI antenna and SCM sensor. The boom is sized to meet the magnetic cleanliness requirements while avoiding innovative design and deep modification of platform electronic or need for massive shielding.
- A secondary structure located on zenith part of the +X wall of the spacecraft hosts the remaining PEP units JDC (PEP_Lo) and JoEE (PEP-Hi) , where they are protected from direct propulsion plumes without interfering with their fields of view;
- Two vaults along central cylinder gathering most of the instruments and platform electronic units in order to provide an efficient shielding against the harsh Jupiter radiation environment, a warm thermal environment (also beneficial for tanks & propulsion hardware) and an ElectroStatic Discharge (ESD) & ElectroMagnetic Cleanliness (EMC) tight cavity.
- A High Gain Antenna (HGA) to download, in both X and Ka band, an average of 1.4Gbits/day of science data, and to ensure TeleMetry (TM)/TeleCommand (TC) links for nominal operations & safe-mode, as well as supporting the Radio Science Experiment. The High Gain Antenna is accommodated to protect equipments from the high solar flux during the hot cruise phase of the interplanetary transfer.
- A solar array with two wings of five panels each for a total surface of 85m² along Y axis providing 730 W at Jupiter, end of life conditions.
- A 425N bi-propellant main engine aligned with the main thrust axis to perform main mission manoeuvres (Jupiter and Ganymede Orbit Insertion) with assistance of 2x4 20N bi-propellant thrusters used for attitude

control and as backup to the main engine in case of failure. 2x6 10 N bi propellant thrusters complement the CPS actuators to provide a pure torque capability about the 3 axes.

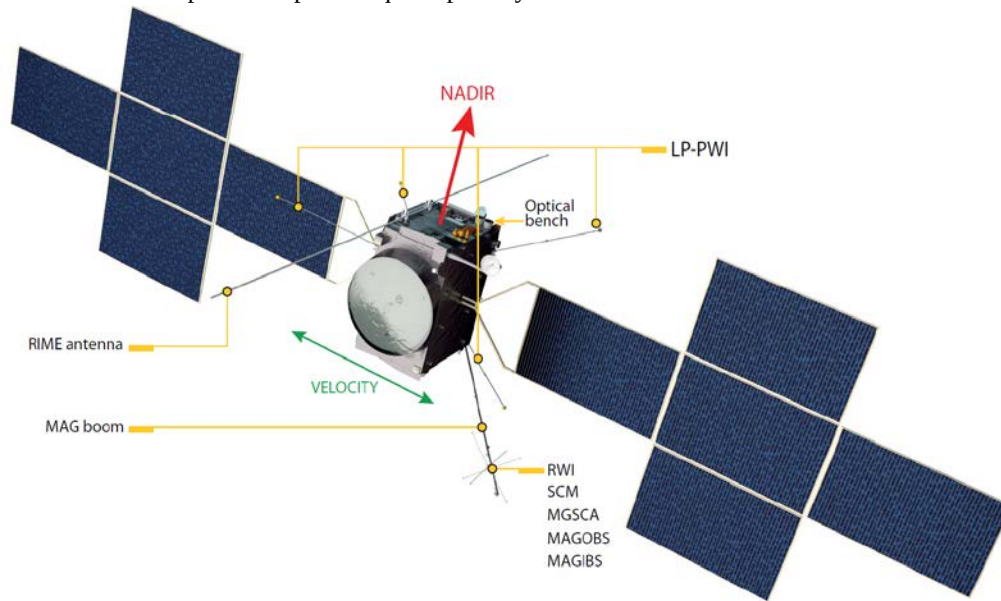


Fig. 3 JUICE spacecraft, all appendages deployed

In addition, several platform equipments are accommodated inside the structure: six propulsion plates accommodated close to the vaults & the central cylinder hosting propulsion sub-assemblies, three pressurant tanks (Helium) & batteries accommodated near the vaults in order to provide additional shielding to electronic units, four reaction wheels accommodation on the Central Cylinder. All the remaining equipments are externally accommodated with especially:

- The deployable appendices including payloads such as RIME radar antenna, J-MAG magnetometer boom and RPWI appendices as well as the steerable Medium Gain Antenna (MGA) used for communication during the “Close to Sun” phase & survival mode.
- The Sun sensors used for the Sun acquisition and the Low Gain Antennas (LGA)s used for communication close to the Earth & during survival mode.

C. Spacecraft electrical architecture

The electrical architecture is built around a central Command and Data Management Unit (CDMU) which gathers functions of a central computer (i.e. command, telemetry and data management, datation and 1 Hz synchronisation management and Failure Detection Isolation and Recovery (FDIR) management) and separate mass memories used respectively for science and housekeeping data storage. A separate Remote Interface Unit (RIU) provides discrete standardized interfaces but ensures also interface adaptation with non-standard platform equipment (propulsion, sun sensors, reaction wheels).

Communications with platform equipment are ensured by two MIL-STD-1553 B data bus. Nominal units are connected on one bus while redundant units are connected to the other, ensuring no common failure point between nominal and redundant communication channels.

Data exchanges with the payload instruments are ensured by SpaceWire links, both for Command and Control (C&C) and for collection of scientific data. SpaceWire links are connected in a network with the routing function allocated on the mass memory boards of the CDMU. The SpaceWire inherent system time distribution and the PUS service 9 are used for time synchronization with the instruments. The SpaceWire network is compatible to the high rate instruments. Inter-instrument communication is established via the SpaceWire network.

The electrical architecture provides full redundancy by its unit and bus architecture with related cross-strapping except for accepted single point failures as e.g. antennas.

The Power Control and Distribution Unit (PCDU) provides autonomous power management (battery charge/discharge, solar arrays power regulation). Switching and protection of all power lines is performed in the

PCDU. All essential (“vital”) loads are protected by Permanently ON (and non-commandable) active current limiters (PO-CL).

The central PCDU uses a Maximum Power Point Tracker (MPPT) for compatibility to the different mission phases and provides a bus voltage of regulated 28V. A dedicated up-converter is included in the PCDU for the 100V supply to the reaction wheels. Energy required during eclipse phases and at any time during the mission the Solar Generator is not sufficient to satisfy the spacecraft power demand is extracted from the Battery Assembly consisting of six Battery Modules and controlled by the Battery Charge and Discharge Module. The power subsystem includes for each wing of the solar generator the release and deployment devices and a one axis Solar Array Drive Mechanism (SADM).

The RF communication relies on X-band for TM/TC and a combination of X- and Ka-band for downlink of scientific data. Low Gain Antennas (LGA) providing omni-spherical coverage are used for near Earth TeleMetry (TM)/TeleCommand (TC) and for emergency TC access. The two axes steerable MGA is foreseen for TM/TC during the inner part of the Cruise, when the HGA is used as a Sun shield, and for Venus Gravity Assist, emergency TM access and radio science during fly-bys in Ka- and also X-band. The fixed HGA is the nominal communication antenna at Jupiter for downlink of scientific data, TM/TC and radio science experiment.

The JMU (JUICE Monitoring Unit) contains the electronics for the evaluation of degradation of sample photovoltaic cells, micro-meteorite damage on solar generator panels, vault internal radiation monitoring and the power supply for the JMC (JUICE monitoring cameras). The cameras are directly connected via SpW to the CDMS. The JMAG sensors can be calibrated by JACS (JUICE Alignment Calibration System), a coil which is powered by the RIU.

The AOCS is based on a combination of Sun Acquisition Sensors (SAS), Inertial Measurement Units (IMU) embedding accelerometers and fiber optic gyros, Star trackers (STR) and Navigation Cameras (NAVCAM) as sensors, and Reaction Wheels (RW), 425 N main engine, 10 N thrusters and 20 N thrusters as actuators.

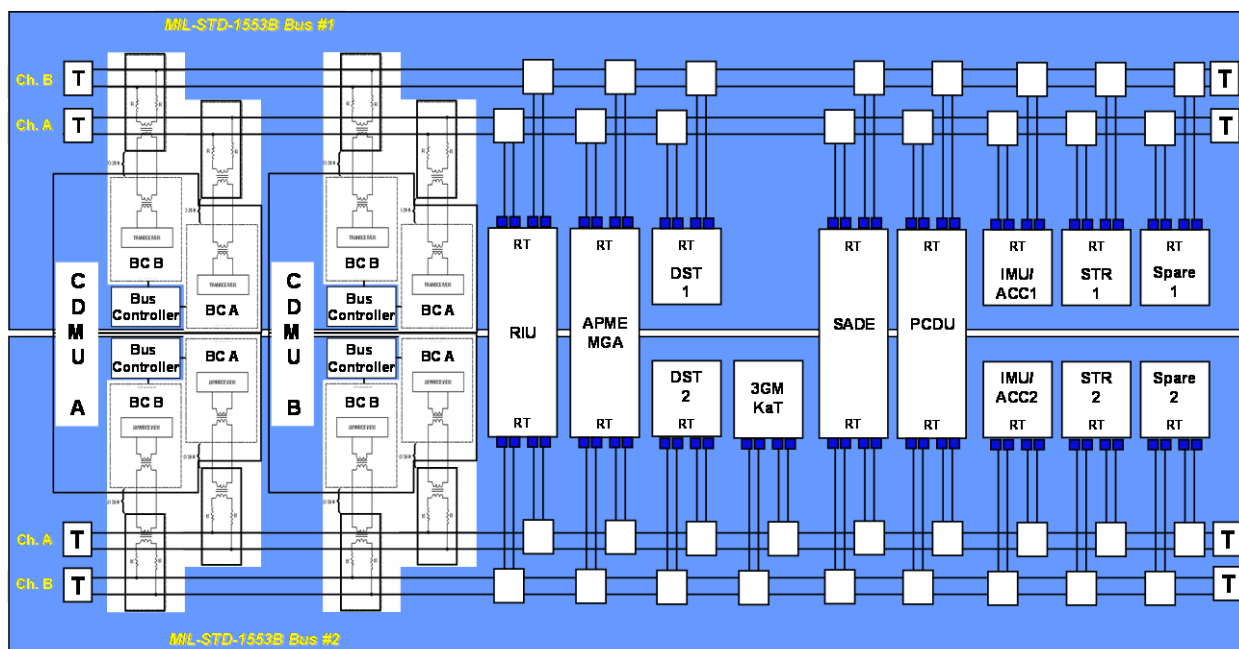


Fig. 4 JUICE platform units architecture around 2 1553-MIL buses

V. Autonomy and robustness

A. Autonomy in nominal situation

1. File Transfer and Mission TimeLine

JUICE nominal science operations are based on a single 8hours ground-station pass every day. The rest of the time, JUICE shall be fully autonomous.

NOTE: In some circumstances, this nominal observation scheme may be adapted: during quiet period of the interplanetary cruise, spacecraft contact is reduced to 8 hours per week, while, in LEOP or in contingency situations, 24/24 coverage may be ensured by ESA ground station network.

JUICE is never operated in direct commanding (except in LEOP), not only because of the round-trip delay (up to 80 minutes) but also to be robust to a spurious ground station outage. Spacecraft operations are split into sequences that shall be either entirely execute or not executed at all. To guarantee the telecommand sequences integrity, they are grouped in a file and the file is uploaded using file transfer (so-called PUS Service 13). This protocol ensures that the telecommands are executed only when the full sequence is confirmed to be received on-board (including verification of the file checksum). This protocol also allows to re-send missing parts of the files (if some where lost during the upload).

For executing operations out-of-visibility, the telecommand sequences are inserted in the Mission Time Line (MTL) with a time-tag. When the pre-programmed time elapses, the telecommand is executed.

ESOC makes intensive use of the combination of File Transfer and Mission Time Line (telecommands to insert operational telecommands in MTL are uploaded by File Transfer) to offer maximum telecommanding reliability (see Fig. 5).

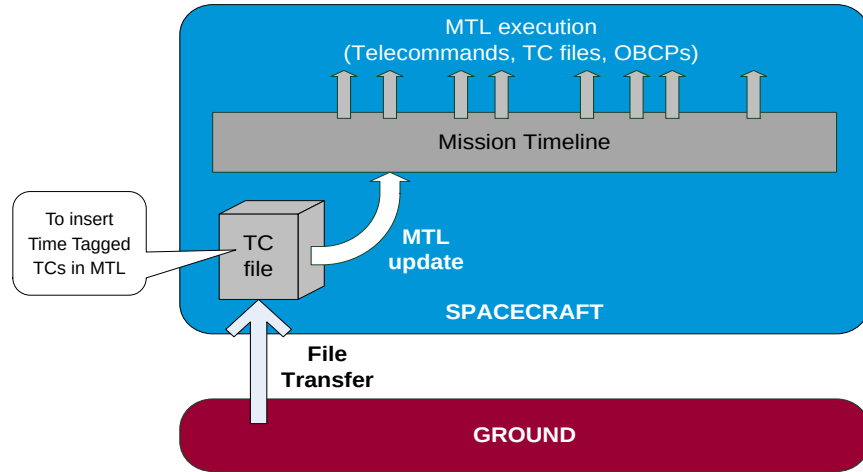


Fig. 5 File transfer and mission time-line to secure spacecraft commanding

2. Autonomy per sub-system

Some sub-systems require few interventions: power management is fully autonomous and ground has only a few parameters to set depending on the mission phase (battery charge level, battery charge/discharge current). However, ground and especially mission planning shall take care to not deplete the batteries to their minimum state of charge (and ensure that batteries are fully recharged before long eclipses). Thermal control is also fully autonomous and ground has only to change some temperature thresholds required for particular operations.

Some sub-systems are fully under ground control like the instruments (observation sessions are programmed by ground – instruments operations are further detailed in section 0) and the communication sub-system (telemetry part). Ground schedules transmission sessions and programs the data rates in X-band and Ka-band (variable as a function of ground station elevation to maximize the daily data return). The telecommand chain of the communication sub-system is fully autonomous.

The Attitude and Orbit Control Sub-system requires a heavy commanding from ground: Flight Dynamics team at ESOC is in charge of preparing and verifying all AOCS commands: sun and planet ephemerides (including Jupiter moon), attitude profiles for science, velocity increment for orbit manoeuvres, navigation camera images acquisitions. When all these data are loaded on-board, the AOCS executes autonomously the commanded attitude

profiles, points automatically the Solar Arrays to the Sun and the steerable Medium Gain Antenna to the Earth and stop thruster and/or main engine firing when commanded velocity increment is achieved. Navigation is principally performed on-ground using radiometric data gathered in dedicated ranging sessions (so-called delta-DOR). Navigation is further improved in the vicinity of the moons by optical images from the NAVCAM processed on ground. This approach is mandatory to navigate in the Jupiter system because Jovian moons ephemerides are not accurately known and fly-bys target a low altitude. However, to improve the spacecraft pointing at closest approach, Airbus implements an autonomous on-board guidance correction based on NAVCAM images.

B. Autonomy in contingency situations

1. Hierarchic Failure Detection, Isolation and Recovery strategy

The on-board failure management has two main objectives:

- (i) The first and most important goal of the FDIR is to ensure the spacecraft integrity, including the platform equipment and the scientific instruments.
- (ii) As far as possible, the FDIR optimizes the mission operational availability.

During most of the mission phases, the two objectives can be separated, and the main goal of the FDIR is to have a “Fail-Safe” action in case of failure, selecting the solutions which ensure the spacecraft integrity.

During critical phases of the mission, the Jupiter Orbit Insertion (JOI) and the Ganymede Orbit Insertion (GOI), the two objectives cannot be distinguished, since a failure during this phase could affect the spacecraft safety and could lead to the loss of the mission. In this case, both objectives have to be ensured, and the goal of the FDIR is to have a “fail-operational” behaviour, enabling the continuation of the operations after the failure. This means that, in case of a single failure before or during the performance of Main Engine Boost Mode (MEBM), the spacecraft is able to achieve the manoeuvre either with the Main Engine (if still operational) or the thrusters.

2. JUICE redundancy and reconfiguration concept

All JUICE platform units are reduded and are operated under one of the following schemes:

- (i) Cold redundancy: the nominal unit is used and the redundant unit remains off.
- (ii) Warm redundancy: the nominal unit is used and the redundant unit is on, ready to take-over in case of nominal unit failure
- (iii) Hot redundancy: both nominal and redundant units are used in the same time.

The usage of each scheme is not static and evolves according to the criticality of the mission phase and other spacecraft constraints (mainly power-related constraints). Warm or hot redundancy will be favored during mission critical phases but induce a higher power consumption and therefore cannot be permanently applied. During science observations, platform power shall be minimized to allocate as much power as possible to instrument operations.

Whenever possible and whenever spacecraft integrity is guaranteed, the FDIR will privilege local reconfiguration (e.g. switch from nominal to redundant unit). Local reconfiguration limits the FDIR action to what is strictly necessary in order to achieve the second objective i.e. minimize the impact on mission availability. In all other cases (i.e. when local reconfiguration is not possible/desirable), JUICE will autonomously switch to Safe Mode. The use of warm redundancy is an “enabler” for local reconfiguration to avoid Safe Mode triggering. A simple example illustrates this concept: JUICE gyroscope requires about 20 seconds after switch-on to provide good measurements. In case of nominal gyroscope failure (and redundant gyroscope is off – cold redundancy), it is considered unsafe to live without gyros for 20 seconds and, in this case, the only possible action is to go in Safe Mode. In critical mission phase, the redundant gyroscope is switched-on (warm redundancy) and is ready to take over nominal gyroscope in case of failure. The swap from nominal to redundant is quasi-immediate and Safe Mode is avoided.

Table 1 provides, for some units, the redundancy and reconfiguration concept according to the mission phase criticality.

Mission Phase		TC reception	TM emission	Instruments	Solar Array pointing	Antenna Pointing
Nominal	Redundancy	Hot	Cold	ON	Cold	Cold
	Reconfiguration	Silent	Local	ON->OFF	Local	Local
Critical	Redundancy	Hot	Cold	OFF	Cold or Hot	Cold
	Reconfiguration	Silent	Local	N/A	Local	Local

Mission Phase		Gyro	STR electronics	STR optical heads	Reaction Wheels
Nominal	Redundancy	Cold	Cold	Hot (3 OH)	Hot (4 RW)
	Reconfiguration	Safe Mode	Local	3->2	4->3
Critical	Redundancy	Warm	Cold	Hot (3 OH)	Hot (4 RW)
	Reconfiguration	Safe Mode	Local	3->2	4->3

Table 1: Example of redundancy and reconfiguration concept

The strategy for critical mission phases is to switch-off all units that are not mandatory for the success of the mission phase, e.g. instruments for JOI or GOI will be switched-off prior to the manoeuvre. The strategy may also change during the mission phase itself, e.g. solar arrays drive electronics are operated in cold redundancy during Normal Mode prior to JOI/GOI but will be operated in hot redundancy during the Main Engine burn (to avoid Main Engine switch-off to reconfigure the Solar Array drive mechanism electronics).

Multiple failures are not considered but reaction to multiple failures may be different depending on the mission phase criticality. For instance:

- Double TM emission failure leads to Safe Mode in nominal mission phase whereas it leads to simply TM emission switch-off in critical phase (TM emission is not a mandatory function for critical manoeuvre execution)
- Multiple wheel failure leads to Safe Mode in nominal mission phase whereas it leads to switch to thruster controlled mode to continue the execution of the critical manoeuvre.

3. Safe Mode concept in Fail-Safe conditions

JUICE Safe Mode is managed by the CDMU. When Safe Mode conditions are detected (by the mean of HW or SW alarms), the processor module will be reconfigured and restarted according to pre-defined reconfiguration sequences (driving the HW selection to be used in Safe Mode). Since the FDIR is active in Safe Mode, it is possible to trigger a (short) series of reconfigurations, i.e. successive Safe Modes exploring different HW configurations aiming at mixing CDMU configurations and other units configurations as illustrated in Table 2.

Reconfiguration level	Processor module	Other Units	Instruments
Nominal	A	A	A or B
First reconfiguration	B	B	OFF
Second reconfiguration	A	B	OFF
Third reconfiguration	B	A	OFF
Fourth reconfiguration	A	A	OFF

Table 2: HW selection in case of successive reconfigurations

At the last Safe Mode reconfiguration, the Safe Guard Memory (holding context data) is no longer used (default SW image parameters are used instead) because the most likely cause for reaching the ultimate Safe Mode is an

error in the context. The FDIR is also disabled at this level to avoid endless loops (i.e. Safe Mode triggering again and again and again).

In Safe Mode, JUICE is automatically put in the safest configuration from power and communication point of view: High Gain Antenna pointed to the Sun (in hot cruise phase) or the Earth (out of hot cruise), MGA pointed to the Earth (in hot cruise phase), solar arrays pointed to the sun with a 70° tilt in hot phases, directly to the Sun out of hot cruise. The parameters required to safely execute the Safe Mode (e.g. Earth ephemerides, HGA pointing options) are stored in the Safe Guard Memory in EEPROM operated in hot redundancy. All these parameters are carefully updated by ground either regularly (ephemerides) or upon mission phase change (e.g. transition from ‘hot cruise’ to ‘cold cruise’). Upon ground request, the Command and Data Management Subsystem allows to re-activate, from SGM, a pre-programmed mission time line, the so-called back-up MTL. In nominal conditions, the back-up MTL is not used, but is part of the fail-operational concept detailed in section 4).

From the AOCS point of view, JUICE Safe Mode starts by a rate reduction using gyroscopes and thrusters, followed by a Sun acquisition (HGA pointed to the sun) with sun sensors, gyroscopes and thrusters. A 3-axis attitude is then acquired using Star Tracker (attitude control is still with thrusters). JUICE is then autonomously pointed to restore communication, either HGA is pointed to the Earth or HGA remains sun-pointed and MGA is steered towards the Earth. Eventually, attitude control is transferred from thrusters to reaction wheel and autonomous wheel off-loading may be triggered from time to time, depending on ground intervention after Safe Mode entry (in the worst case, Safe Mode entered at the start of a conjunction phase, JUICE shall survive without ground contact for about 30 days). If authorized by the ground, it is possible to by-pass the sun acquisition phase and directly acquire a 3-axis attitude with Star Tracker to speed-up the Safe Mode convergence. This capability will be intended to be used in fail-operational conditions (see more details in section 4).

In case Star Tracker does not provide valid measurement (or if the last reconfiguration level is reached), JUICE enters in Survival Mode. This specific mode is called “Earth strobing” (inherited from Rosetta). The HGA remains sun pointed, the MGA is moved to a fixed position and the spacecraft rotates slowly such that the MGA beam intercepts the earth at each rotation. This will allow the ground station to catch the RF signal carrier and to execute the procedure “recovery from Survival Mode” which requires, at the beginning, to send commands in the blind (i) stop spacecraft rotation, (ii) reduce attitude control limit cycle to improve pointing performance and (iii) restore a minimum of telemetry to assess the spacecraft state. The Survival Mode recovery being quite tricky, the AOCS continues trying Star Tracker acquisitions, and as soon as valid measurements are recovered, the usual Safe Mode is continued – recovery from usual Safe Mode being far easier.

4. Fail-operational concept

Airbus carefully analyzed the time constraints and the driving requirements related to the critical manoeuvres such as the Jupiter Orbit Insertion. The analysis showed that the propellant penalty is not significant in case the manoeuvre is resumed within 15 minutes from an interruption due to a critical failure. Different fail-operational concepts were traded with respect to complexity in the implementation and the related validation. The selected fail-operational strategy relies on:

- (i) A Safe Mode as described in section 3, with active damping of the Solar Arrays flexible mode along the thrust axis and direct transition from rate reduction phase to 3-axis attitude acquisition, and use of a ground commanded attitude profile i.e. in this case, the attitude recovered by the Spacecraft in Safe Mode is not a “earth-pointed” attitude but an insertion attitude. This is made possible thanks to the use of the back-up Mission Time Line and dedicated attitude guidance segments stored in SGM.
- (ii) Once the Safe Mode is converged (with thruster only – to speed-up the convergence), the critical manoeuvre is restarted as soon as possible.

The major advantage of this approach is to rely on existing features and therefore minimizes the development and validation risks. It offers also a flexibility that can be used by ESOC to customize the strategy.

VI. Instrument file-based operations

A. Overview of instrument data management

The instruments, the science mass memory and the on-board computer are interconnected thanks to the spacewire router. Since for most of the instruments the nominal and redundant spacewire interfaces are not internally cross-strapped, they are connected to the router function via a cross-trap function (i.e. cross-point switches). This cross-strap function allows routing the nominal or redundant spacewire link from each instrument to either routing function A or routing function B. Each data stream is identified by a logical address and the routing function is configured to route the corresponding data flow from the emitter to the receiver.

NOTE: Fig. 6 is simplified. There are in total 13 units (with dual spacewire interface) connected to 2x13 cross-points and there are 4 additional links directly connecting each SSMM to each OBC (not represented here). This additional connection does not go through the router and is used for mass memory command/control by the OBC;

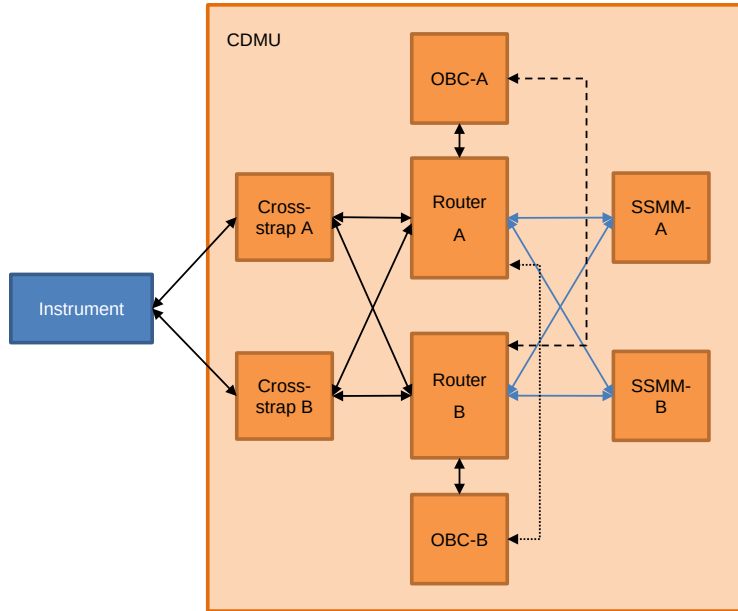


Fig. 6 On-board computer, Mass Memory and instrument inter-connection overview

The data routing over the spacewire network is based on logical addresses. An allocation of logical address to each instrument is performed as follows:

- One logical address for instrument commanding (from OBC to instrument),
- One logical address for instrument ‘non-science’ telemetry (from instrument to OBC),
- 2 to 8 logical addresses for instrument science telemetry (from instrument to SSMM)

In addition, a dedicated logical address is allocated to a specific case of instrument real-time inter-communication and two dedicated logical addresses between OBC and SSMM for test and monitoring purpose.

The router is statically configured to route spacewire packets to the correct physical port according to the logical address (provided in the packet header). The router features a packet distribution mode that allows routing packet to multiple ports. This capability will be used on Juice to operate the 2 SSMMs in mirror mode (hot-redundancy) for critical science observations (namely the two Europa fly-bys which represents unique opportunities).

B. Instrument monitoring and control

All instruments connected to the router are PUS-based and therefore, the CDMS only acts as a router of the instrument TM/TC to/from ground. Non-science telemetry (from instrument to OBC) allows implementing instrument monitoring by the platform or by ground:

- For ground observability, non-science telemetry is recorded in packet stores in the OBC Mass Memory (physically separated from the Science Mass Memory).
- For platform monitoring (illustrated in Fig. 7, relevant events or house-keeping telemetry parameters are monitored and are associated to recovery action initiated by the platform. Since we anticipate that the

instrument monitoring will require flexibility both on-ground (during the development phase) and in-flight, all instrument monitorings are implemented by using standard PUS Service 12 and 19. The recovery action is implemented either via a single telecommand, a telecommand sequence or on on-board control procedure (OBCP).

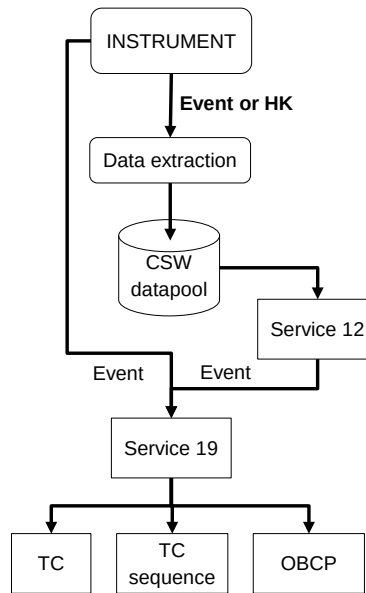


Fig. 7 Instrument monitoring by Central Software

C. Science acquisition

Once the router is configured by the ground, the science data acquisition is fully autonomous. Each spacewire destination logical address from instrument to SSMM is associated to a directory in the Science Mass Memory. The Science Mass Memory records into files (in each directory) all incoming spacewire packets (after removing the 4bytes spacewire header). Contrarily to the packet stores, the science data stored in SSMM files do not need to respect PUS standard. Some instruments have chosen to use PUS format, some other, for efficiency reason (and especially in the case data is compressed by the instruments before been sent to the mass memory), have chosen a “private” format. Directories have also a maximum size: when this maximum size is reached, the last recorded file is closed and the incoming data are discarded.

The Science Mass Memory automatically closes files (and continues recording data in a new file) upon predefined (but configurable criteria): (i) maximum file size is reached or (ii) maximum time since first recording is reached. These criteria, attached to each directory, will be statically defined but may be changed by ground) depending on the mission phase. These criteria work well for instruments producing a constant stream of science data, with no special needs regarding the relationship between observation and generated files.

On top of the simple file switching criteria mentioned above, the instruments are also able to control precisely the time of the file switch to make sure that one observation is stored in a file and the next observation is stored in the next file. This functionality is supported by: (i) the capability to switch files by telecommand, (ii) the capability of the CSW to react on an event issued by the instrument, (iii) the so called “flip-flop” mechanism where the instrument toggles between two destination directories while producing the science data stream.

The file switch via telecommand is a simple solution based on the mission time-line as illustrated in Fig. 8. This approach requires determining accurately when the last packet of the image #1 (or observation #1) is recorded in file#1 before executing the command to switch file and then start the second observation. This strategy can work for instruments where the individual observations can be scheduled well in advance and the time between observations is compatible with the MTL (and telecommand execution) tolerances.

To implement a more autonomous file switching, the CSW also allows the instrument to issue an event to notify the end of the current observation. The CSW then switches file by telecommand using standard PUS Service 19. This principle is illustrated in Fig. 9. This mechanism does not require a planning in advance of the file switching, but it is not suitable if the delay between subsequent observations is below the time needed by the CSW to process the event (e.g. a few seconds)

To eventually answer to the need of successive observation stored in successive files without a significant delay between observations, A so-called “flip-flop” mechanism is implemented. This mechanism requires allocating two logical addresses for one instrument data stream. At the end of observation #1 stored in LA1/File#1, the instrument sends the packets corresponding to observation #2 to LA2. Fig. 10 shows that the delay between two successive observations is minimal (and only limited by the instrument capability to do successive observations). While the second observation is recorded in LA2, there is “plenty of time” to switch the file in LA1 (using the event mechanism described above) before starting the third observation (back) on LA1 when the second observation (on LA2) is completed.

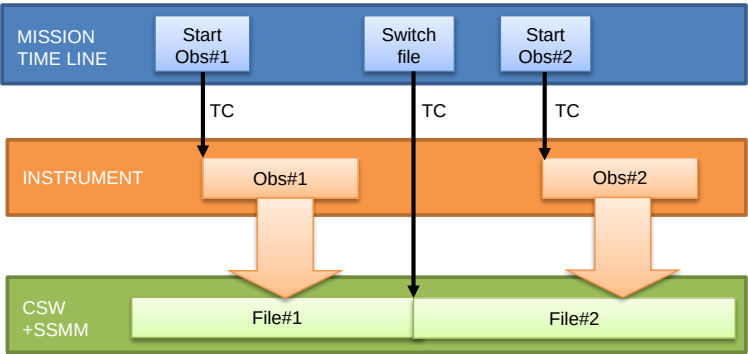


Fig. 8 Synchronizing observation and file switching (by TC in MTL)

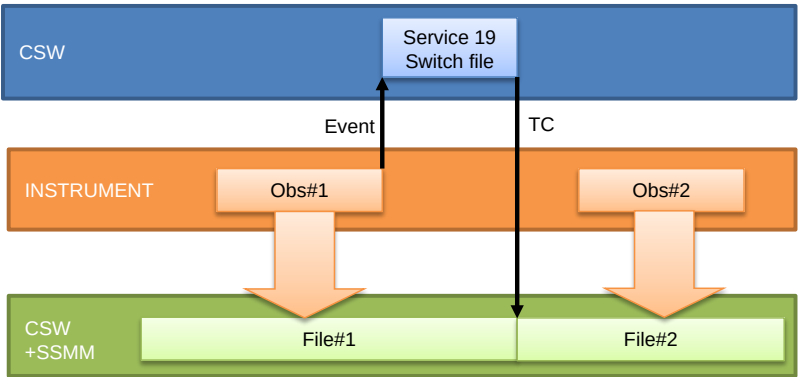


Fig. 9 Synchronizing observation and file switching (by event)

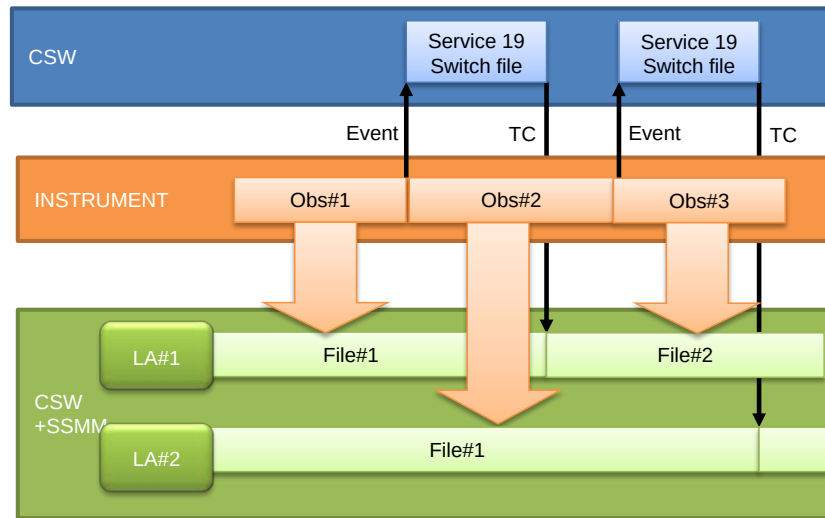


Fig. 10 Logical addresses in flip-flop and file switching

D. Science data downlink

1. CFDP

Only closed files are candidate to downlink. Each individual file of the SSMM is transferred to ground using the CCSDS File Delivery Protocol (CFDP). Each file to be downlinked is split into small pieces (called Protocol Data Unit – PDU) to be downlinked in individual CCSDS telemetry packets.

The file transmission is started by Spacecraft by sending the file meta-data (transaction identifier, file name, file size and number of PDUs to be transmitted) and then, each individual PDU is emitted to the ground. At the end of this first transmission of the file, the CSW emits an “End of File” (EOF) message to indicate that the file transmission has been completed. The EOF is acknowledged by the ground segment.

From this point onwards, ground can reconstruct the file and check that all expected PDUs have been received. If not, ground can request, by telecommand, the missing PDUs (NAK) until the file transfer is completed and the file is entirely received on-ground. Once the complete reception is confirmed, ground will send an acknowledgement to the spacecraft to terminate the CFDP transaction for this file and possibly delete it from the SSMM.

Note that, depending on the file size and on the transmission quality, the complete reception of a file may take quite a long time. For example, consider a file of 10 SSMM memory blocks i.e. 20 Mbytes. The PDU size is 512 bytes which leads to ~41000 PDUs to be transmitted in ~20500 frames. If one assumes 1% frame loss, 205 frames will have to be retransmitted and among them, still 2 frames will be wrong and will require a second retransmission.

This protocol secures the file transmission but increases the file life time on-board and therefore the memory occupancy. The protocol does not address

- (i) The transport layer: it has been agreed with ESOC that the PDU would be transported using the usual TM (CCSDS format) and telecommand (PUS format).
- (ii) The file selection: the protocol explains how to downlink a file, not how to choose which file to downlink among all the files recorded on-board.

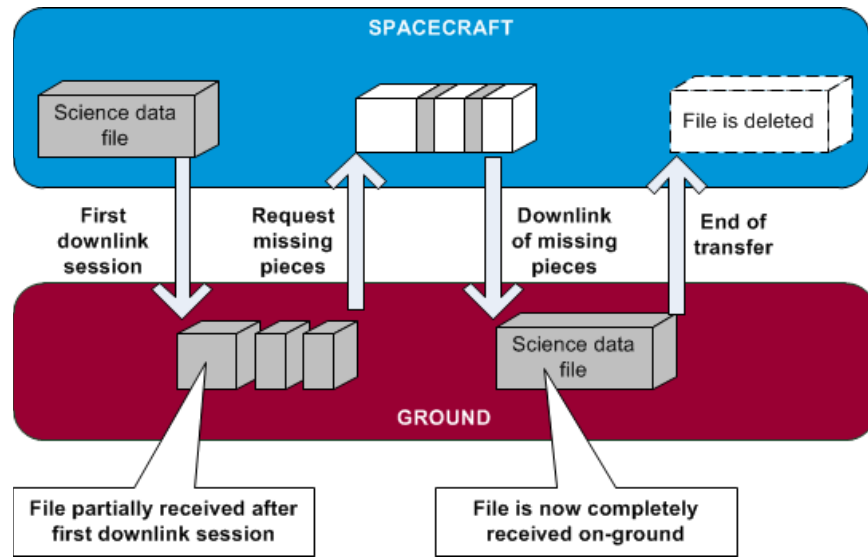


Fig. 11 CCSDS File Delivery protocol principle

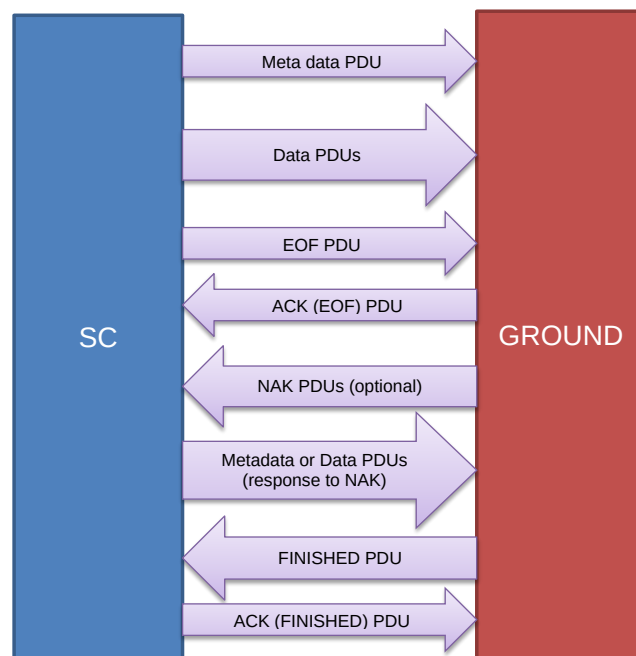


Fig. 12 CCSDS File Delivery protocol exchanges

2. File Selection

The file selection could be completely manual i.e. done by ground telecommand. But we have seen that the file switching (and therefore file naming) is fully autonomous on-board, hence selecting files to be downlinked one by one was not considered as a viable option. Following ESOC requirements, a priority mechanism at directory level has been implemented, such that file selection for downlink is managed in a fully autonomous manner. For this purpose, each directory features three attributes (managed by ground):

- the downlink RF path (X or Ka),
- the eligibility for downlink,
- and the downlink priority.

As soon as a file is closed, it becomes candidate for downlink (if its directory is eligible for downlink). The file created in a directory inherits the RF band for downlink from its parent directory at the time of downlink starts. Within one RF band, directories of the same priority level are polled one after the other in a round-robin manner. Within one directory, the files are downlinked from the older to the most recent.

In case a file is not completely received on ground during its initial transmission, the CCSDS File Delivery Protocol foresees that the ground can request again the missing data segment (protocol data unit). Hence, the Central Software receives from the ground PDU re-transmission requests (so-called NACKs). The PDU re-transmissions have to be prioritized with respect to file transmission. The following principles are applied:

- The data retransmission is given maximum priority (zero).
- Re-transmission is systematically performed in the same band (X or Ka) as the original transmission.

Functionally speaking, the management of the CFDP NACKs can be seen as adding a dedicated directory per RF band. When CFDP NACK are received from ground (i.e. instruction to the CSW to retransmit missing PDUs), the corresponding PDUs are put in this directory priority zero (maximum priority), after the PDUs (if any) already present. With this approach, the PDUs are re-transmitted in the order of reception of the NACK from the ground.

Fig. 13 illustrates how the downlink is managed in one RF-band (from the functional point of view) with three directories. Directory A and B have the same priority level while directory C has a lower priority. The last file in each directory (in grey) is still open and is (possibly) recording data coming from the instruments.

The PDU directory is a virtual directory of maximum priority which stores the PDUs to be retransmitted (when the corresponding CFDP NACK telecommand is received from ground).

When downlink starts, it first looks in the highest priority directory and downlinks the first object (here PDU1) present in this directory. When PDU1 downlink is completed, the downlink priority management polls again the directory highest priority and downlinks PDU2 and then PDU3.

When there is nothing left to downlink in the PDU directory, it goes to the directory of next priority. Because directory A and B have the same priority, the downlink priority management will select alternatively files from directory A and directory B. Hence, the downlink order will be (in this example) A1, B1, A2, B2, A3, B3, A4.

Once there is nothing left to downlink at priority level #1, the downlink management goes to priority level #2 and applies the same mechanism. The files C1, C2 and C3 are thus downlinked in this order.

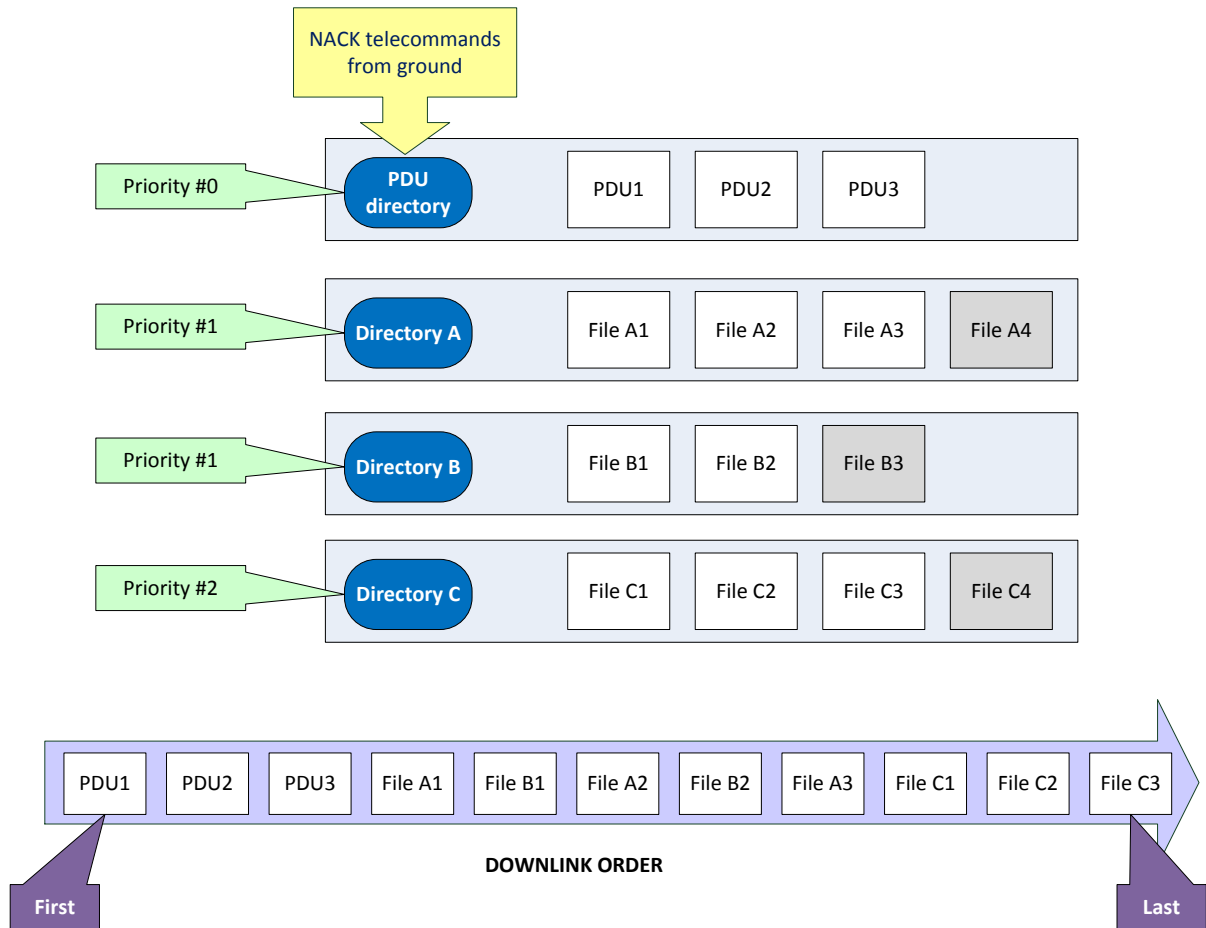


Fig. 13 Downlink priority management (separate for X-band and Ka-band)

For the sake of simplicity, Fig. 13 assumes that NACKs have been received before downlink and that open files remain open during downlink. But the principle described above can be easily extended in case new NACKs would be received during downlink or in case files would be closed. For instance:

- If the ground sends a request to retransmit a new PDU (PDU4), during downlink of file A2, this new PDU will be inserted in the downlink before the downlink of file B2. Hence the sequence will be A2, PDU4, B2 (and will continue as above).
- If the file A4 is closed (and therefore becomes candidate for downlink) during the downlink of file C1, file A4 will have precedence and therefore, file A4 will be downlinked before file C2. The new sequence will be C1, A4, C2 (and will continue as above).

3. Selective downlink

Some instruments may generate a lot of data but it is possible that not all the acquired data have a high scientific interest. The file selection described above can be used to implement “selective downlink” which will allow the scientists to select data of interest (and therefore optimize the downlink bandwidth usage). Let consider the simple example of an instrument generating two science data streams (see Fig. 14):

- The first stream has a small data rate and corresponds to the data selector (for instance low resolution images). It represents a small amount of data that is stored in a dedicated directory (or logical address) and that can be downloaded (every day) with a high priority.
- The second stream has a high data rate and corresponds to the data to be selected (for instance high resolution images). It represents a large amount of data stored in a dedicated directory (or logical address) which is not candidate for downlink.

Once the data selector has been analyzed by the instrument team, the instrument team is in a position to define which of the “data to be selected” are effectively of interest and are to be downlinked. This can be done either by deleting the files that are not of interest or by moving the files of interest in a directory that is candidate for downlink.

Obviously, selective downlink cannot be implemented in an autonomous manner since it requires a feedback from preliminary analysis of the data selector (by the instrument team) and a manual selection of files of interest among the data to be selected. The overall turn-around time is estimated to 1 month, increasing again the life time of the (large) files in the Science Mass Memory and memory occupancy.

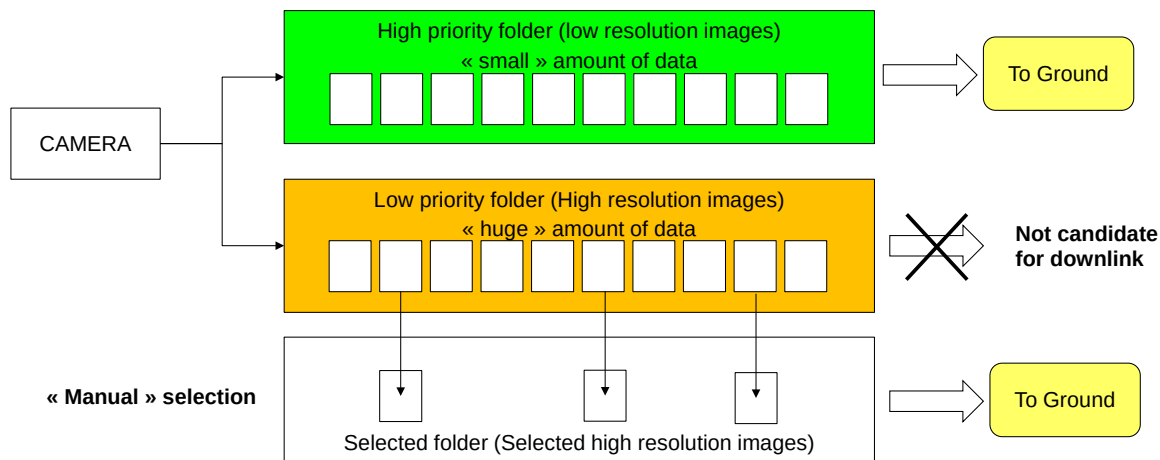


Fig. 14 Downlink priority management (separate for X-band and Ka-band)

VII. Conclusion

This paper provides an overview of JUICE operations, with a focus on the mandatory autonomy level required by interplanetary missions, in both nominal and contingency situations. The brand new file-based operation concept is presented. JUICE experience gained in this domain will serve as a basis for extending this concept to Earth Observation spacecraft (on-going study in Airbus for ESOC).

The next steps of JUICE development are the start of the Spacecraft Engineering Model over the summer 2018 and the Critical Design Review to be concluded by March 2019. Rendez-vous is taken on May 20th 2022 for JUICE launch from Kourou.

VIII. Acknowledgments

The authors would like to thank ESA and Airbus representative for supporting and authorizing the publication of this paper, in particular Giuseppe Sarri, JUICE project manager at ESA, Andrea Accomazzo, JUICE ground segment manager at ESOC and Didier Radola, JUICE project manager in Airbus.